

Notes: Baker, Johnson, and Kueng (JFE, 2024)

Financial Returns to Household Inventory Management

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1 Big picture

- **Question.** Do households earn meaningful financial returns from managing inventories of consumer goods, and should those inventories be viewed as part of household portfolios?
- **Core idea.** Households can use working capital to stockpile consumer goods when prices are low or when bulk purchases are cheaper. This converts cash into stored consumer goods and can generate high effective returns.
- **Main measurement contribution.** The paper constructs a household-level inventory measure from NielsenIQ scanner data by comparing the timing of purchases with the timing of inferred consumption.
- **Portfolio interpretation.** Consumer goods inventory is a nonfinancial asset. It can be compared to financial wealth because both are uses of household working capital.
- **Empirical fact.** Households hold large amounts of inventory. The mean inventory value in the baseline measure is about \$725, and the median inventory-to-spending ratio is roughly 0.20, or 2.4 months of spending.
- **Heterogeneity.** Inventory is especially important for low-income households. In the bottom income quintile, inventory can be a large share of total portfolio wealth; in higher income quintiles, financial assets dominate.
- **Validation.** Product shelf life predicts household inventory holdings, and households reduce spending substantially before moving, consistent with drawing down stored goods before a costly relocation.
- **Model.** The paper builds a household inventory model with two savings channels: buying on temporary deals and buying larger package sizes at lower unit prices. A working capital constraint limits how much inventory can be held.
- **Main return result.** The model implies very high marginal returns at low inventory levels and still high average returns at empirically relevant levels of working capital. The returns are especially important for low-income and low-wealth households.
- **Robustness/validation.** The data show positive relationships between in-store savings and inventory holdings, and the model and data generate similar signs for deal savings, bulk savings, trip intervals, and working capital.
- **Economic takeaway.** Household inventory management can act like a high-return, illiquid investment. This helps explain why some households may rationally devote scarce liquidity to working capital rather than financial assets.

2 Data and measurement

2.1 NielsenIQ household scanner data

- The paper uses NielsenIQ consumer panel data, where households scan purchases over time.
- The data record purchase dates, quantities, prices, UPC/product group identifiers, store type, household demographics, and household shopping trips.
- The core sample covers 2013–2014 for the main inventory measurement, while additional validation exercises use a longer NielsenIQ panel.
- Prices and package sizes are supplemented with NielsenIQ retail price data, which allow the authors to measure temporary deals and bulk discounts.

Interpretation: Scanner data are useful because they observe the timing and prices of purchases. They do not observe consumption directly, so inventory must be inferred.

2.2 Inventory measurement

For household h , product group g , and year y , the paper infers the minimum initial inventory needed to keep inventory nonnegative over the year:

$$I(0)_{y,h,g} = -\min_j I(t_j)_{y,h,g}.$$

The average household inventory is then

$$\bar{I}_h = \frac{1}{2} \sum_{y=2013}^{2014} \sum_g \bar{I}_{y,h,g}.$$

- The method assumes a constant rate of consumption between purchases.
- Inventory naturally excludes goods not observed in the scanner data, especially many durable goods.
- Inventory is reported both in dollars and as a share of annual spending on NielsenIQ-covered goods.

Interpretation: The inventory measure asks: how much inventory must the household have held at the beginning of the period for its observed purchase pattern to be feasible?

2.3 Portfolio measurement using the SCF

- The paper uses the Survey of Consumer Finances to place inventory beside financial assets.
- Household inventory is imputed to SCF households using characteristics observable in both NielsenIQ and the SCF.
- Financial assets include checking accounts, savings accounts, certificates of deposit, money market accounts, bonds, stocks, and mutual funds.
- The inventory portfolio share is:

$$\frac{\text{Inventory}_h}{\text{Financial Assets}_h + \text{Inventory}_h}.$$

Interpretation: This converts grocery inventory from a consumption object into a portfolio object. For many low-income households, inventory is a substantial part of the balance sheet.

2.4 Shelf life and perishability

- Product modules are assigned a usable life in months, based on food safety and product-storage information.
- Product groups are divided into four perishability groups.
- Less perishable goods can be stored longer and therefore should generate higher inventory ratios.

Interpretation: Shelf life is central because it determines whether stockpiling is physically feasible. The empirical relationship between shelf life and inventories validates the measurement strategy.

2.5 Moving events as validation

- Moving is costly when households hold large stocks of consumer goods.
- If measured inventories are real, households should run down inventories before moving.
- The paper identifies moves using changes in household 3-digit ZIP codes.
- Spending falls substantially before moving and then returns to normal after the move.

Interpretation: The mover test is a behavioral validation exercise. Households behave as if they have inventories that are costly to transport.

3 Model and empirical specifications

3.1 Household objective

The household seeks to supply an exogenous consumption flow at minimum cost subject to a working-capital constraint. A trip has a fixed cost k , goods depreciate with perishability δ_l , and purchases can exploit deals or bulk discounts.

Interpretation: The model is not about increasing utility by consuming more goods. It is about reducing the cost of supplying a given consumption stream.

3.2 Stockpiling rule

Let p_f denote the full price and p_d the deal price. Let s_d be the stockpiling target after observing a deal. If current inventory is q , the household's order quantity is

$$q^o = \begin{cases} s_d + 1 - q, & q \leq s_d \text{ and } p = p_d, \\ 1, & q = 0 \text{ and } p = p_f, \\ 0, & \text{otherwise.} \end{cases}$$

Interpretation: At full price, the household only replenishes enough to avoid running out. At a deal price, it buys additional inventory up to the stockpiling target.

3.3 Expected price and deal share

The steady-state share of transactions at the deal price is

$$d = \frac{x}{x + (1 - x)^{s_d + 1}},$$

where x is the probability of a sale. The long-run expected price conditional on the strategy can be written as

$$\bar{p} = \frac{p_d \mathbb{E}[q^o \mid p = p_d] d + p_f (1 - d)}{\mathbb{E}[q^o \mid p = p_d] d + (1 - d)}.$$

Interpretation: A higher stockpiling target raises the probability that purchases occur when the price is discounted, but it requires more inventory.

3.4 Working capital constraint

The household chooses stockpiling strategies $\{s_{l,d}\}_l$ and trip interval Δ subject to a maximum amount of working capital:

$$\sum_l [P_l(\Delta, s_{l,d}) Q_l(\Delta) + I_l(s_{l,d}) Q_l(\Delta) p_{l,d}] \leq \bar{I}.$$

Interpretation: Working capital is used both to buy inventory and to hold cash for the next shopping trip. This is why the returns are not simply retail discounts; they are returns to scarce household liquidity.

3.5 Cost function and marginal return

The model's cost of providing the consumption flow is

$$V(\bar{I}) = \min_{\Delta, \{s_{l,d}\}_l} \frac{k + \sum_l P_l(\Delta, s_{l,d}) Q_l(\Delta)}{\Delta},$$

subject to the working-capital and shelf-life constraints.

The marginal net return to household inventory management is

$$r_I(\bar{I}) = -V'(\bar{I}).$$

Interpretation: If an additional dollar of working capital lowers the cost of household consumption, that cost reduction is the financial return to inventory management.

3.6 Portfolio choice with household working capital

The household's end-of-period wealth problem can be written as

$$\max_{\lambda_I, \lambda_f, \lambda_m} \mathbb{E}U((1 + \tilde{r}_p)w),$$

where

$$\tilde{r}_p = \frac{1}{w} \int_0^{\lambda_I w} r_I(x) dx + \lambda_f r_f + \lambda_m \tilde{r}_m, \quad 1 = \lambda_I + \lambda_f + \lambda_m.$$

Interpretation: Since the inventory return is decreasing in working capital, low-wealth households may optimally hold a large share of wealth in inventory before investing much in financial markets.

3.7 Gross return regression

To estimate in-store savings directly from the data, the paper uses relationships such as

$$\frac{\text{Dollar in-store savings}_h}{\text{Base spending}_h} = a + b_1 \frac{\text{Average base inventory}_h}{\text{Base spending}_h} + b_2 \text{Trip interval}_h + e_h.$$

Interpretation: This regression asks whether households with more inventory achieve more in-store savings, holding trip frequency fixed.

4 Identification and empirical design

4.1 Measurement credibility

- Shelf life is external to household choices and should be mechanically related to feasible storage.
- The shelf-life regressions show that more durable products are held in larger inventory ratios.
- The moving-event analysis shows that households reduce spending before moving, consistent with inventory drawdown.

Interpretation: The paper cannot observe cupboards directly, so it builds a set of indirect validation tests. The shelf-life and mover results support the idea that the inferred inventory is real.

4.2 Cross-sectional heterogeneity

- Inventory ratios vary with age, household composition, income, housing characteristics, and store choice.
- Perishability and store choice have especially high predictive importance.
- The results are mostly descriptive rather than causal.

Interpretation: The heterogeneity results help describe who holds inventory and why, but the authors are careful not to overinterpret them as causal estimates.

4.3 Model identification

- Perishability groups are calibrated using shelf life and expenditure shares.
- Deal probabilities and deal prices are matched to NielsenIQ retail price moments.
- Bulk discount functions are calibrated to pack-size/unit-price relationships.
- Trip fixed costs are chosen to match the average shopping interval.

Interpretation: The model is disciplined by observed shopping behavior, price variation, and product durability, then used to infer financial returns to the unobserved working-capital choice.

4.4 Return validation

- Model-implied relationships between savings and inventory are compared with data relationships.
- The paper separately studies deal savings and bulk savings.
- Movers provide another source of variation: households reduce deal transactions before and after a move, consistent with lower stockpiling.

Interpretation: The paper's return estimates rely on a calibrated model, but they are checked against several reduced-form patterns in the scanner data.

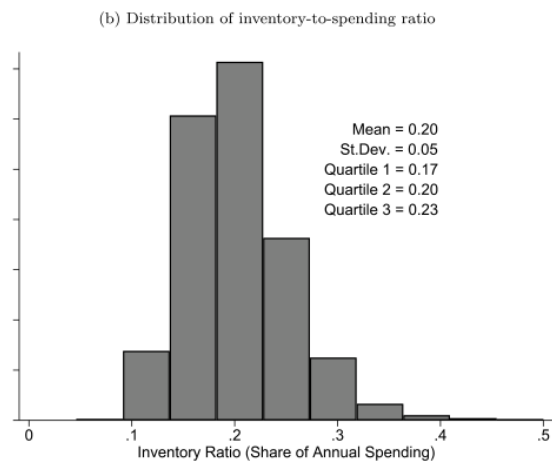
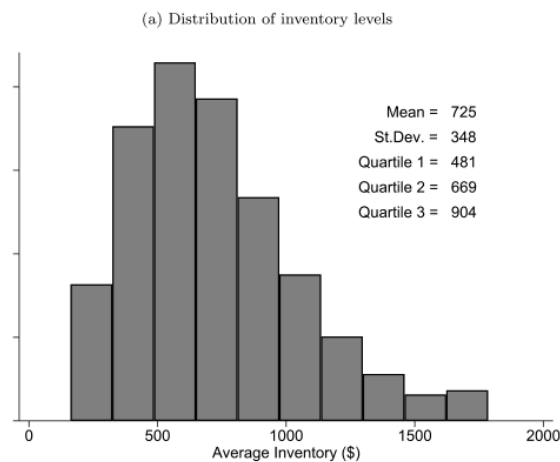
4.5 What the paper cannot fully prove

- It does not observe household consumption directly.
- It does not observe complete household balance sheets, borrowing costs, or credit constraints in the scanner data.
- Inventory returns may also reflect precautionary liquidity motives, storage constraints, preferences, or time costs that are difficult to observe.
- The model abstracts from many household-level frictions such as product waste, storage heterogeneity, and borrowing constraints.

Interpretation: The main contribution is not a fully causal estimate of all household inventory motives. It is a measurement and model-based argument that inventory management can generate economically large financial returns.

5 Tables and figures

5.1 Figure 1 + Figure 2: Inventory as an asset



1. Observed consumer goods inventory.
: Panel (a) plots how the average 2013-2014 inventory level $\bar{I}_h$ varies across households in the NCP. Average inventory is plotted up to the 99th percentile. Summary statistics reported in the top right corner are computed using

Figure 1: observed consumer goods inventory

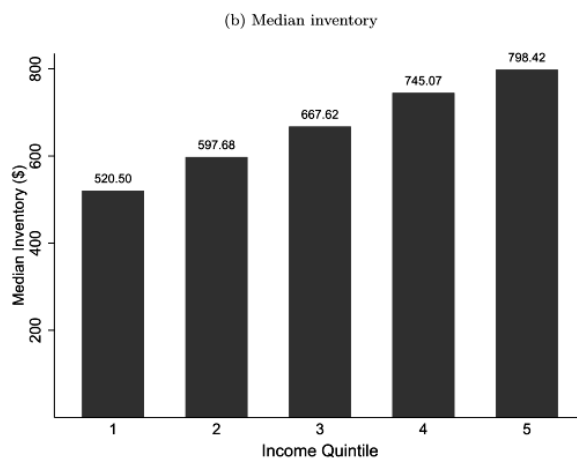
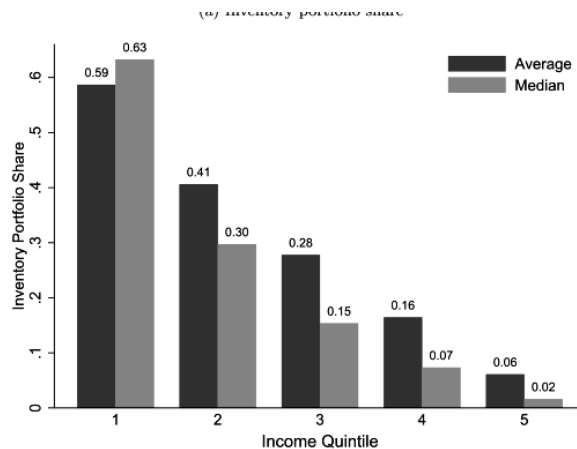


Figure 2: inventory portfolio share by income

Read:

- Figure 1 shows that the mean inventory value is about \$725, and the mean inventory-to-spending ratio is 0.20.
- The median inventory-to-spending ratio is also about 0.20, or 2.4 months of annual spending.
- Figure 2 shows that inventory is a large portfolio share for low-income households.
- In the lowest income quintile, inventory is around 59% of the portfolio on average and 63% at the median.
- Inventory dollar value rises with income, but the inventory portfolio share falls because financial assets rise much faster.

Economic message: Consumer goods inventory is economically large and especially important for households with limited financial assets.

5.2 Table 1 + Figure 3: Validating inventory using durability and moves

Table 1
Validation: relationship between durability and inventory ratio.

	(1)	(2)	(3)	(4)	(5)
Sheff Life (Months)	0.128*** (0.001)	0.300*** (0.002)	0.481*** (0.002)	0.392*** (0.004)	
Sheff Life Squared	-0.004*** (0.000)	-0.004*** (0.000)	-0.004*** (0.000)	-0.004*** (0.000)	
Avg. # Days Between Trips				0.467*** (0.008)	
Group Sheff Life ≤ 58 months					-0.239** (0.027)
Group Sheff Life > 6 months					0.234** (0.027)
Household FE			X		X
Number of Observations	5,578,528	5,578,528	5,578,527	5,535,390	5,578,528
Adjusted R ²	0.07	0.09	0.13	0.15	0.125

Notes: This table combines data from the NCP over 2013 and 2014 and the FSIS. We estimate variations on the following regression specification, where h indexes households and g indexes NielsenIQ product groups:

$$\text{Inventory Ratio}_{i,t} = b_0 + b_1 \text{Shelf Life}_i + b'_2 X_{it} + e_{i,t}.$$

Inventory Ratio_{g,t} is the ratio of household inventory to annual spending in product group *g*, multiplied by 100. Columns 3 and 5 include household fixed effects. Standard errors are clustered by household. Regressions are weighted, using NielsenQ sampling weights multiplied by total product group expenditures. * $p < .1$, ** $p < .05$, *** $p < .01$.

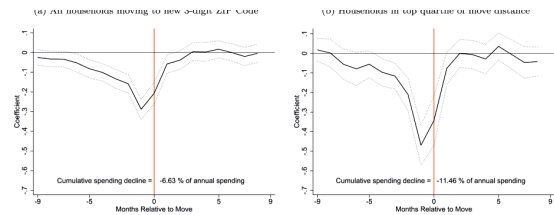


Figure 3: spending around move dates

Table 1: shelf life and inventory ratios

Read:

- Table 1 shows a positive relationship between shelf life and the inventory ratio.
- A longer shelf life raises the inventory ratio, while the squared term implies concavity.
- Household fixed effects do not remove the relationship, suggesting that the pattern is not only cross-household sorting.
- Figure 3 shows that households reduce spending before moving.
- The cumulative spending decline is about 6.6% of annual spending for all movers and about 11.5% for long-distance movers.

Economic message: Households hold more inventory in storable goods and run down inventory before moves. Both patterns validate the inventory measure.

5.3 Table 2 + Figure 4 + Figure 5: Predictors of inventory holdings

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Table 2: factors correlated with inventory ratios

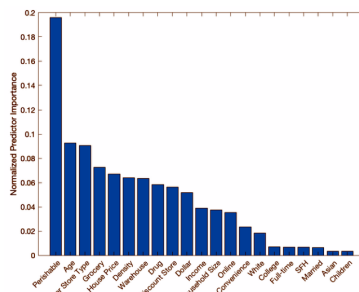


Figure 4: predictor importance

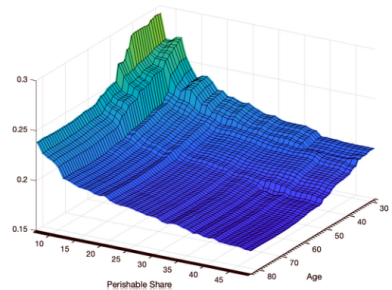


Figure 5: age and perishability

Read:

- Table 2 shows that age, marital status, working status, race, income, housing variables, perishability, and store choice all correlate with inventory ratios.
- Perishable share is strongly negatively associated with inventory holdings.
- Figure 4 shows that perishability is the most important predictor in the machine-learning model.
- Age is the second most important predictor, followed by store type, house price, density, and household size.
- Figure 5 shows that inventory ratios are higher for older households and lower when purchases are more perishable.

Economic message: Inventory holdings reflect both product technology and household shopping/storage circumstances.

5.4 Figure 6 + Figure 7: Heterogeneity and model mechanics

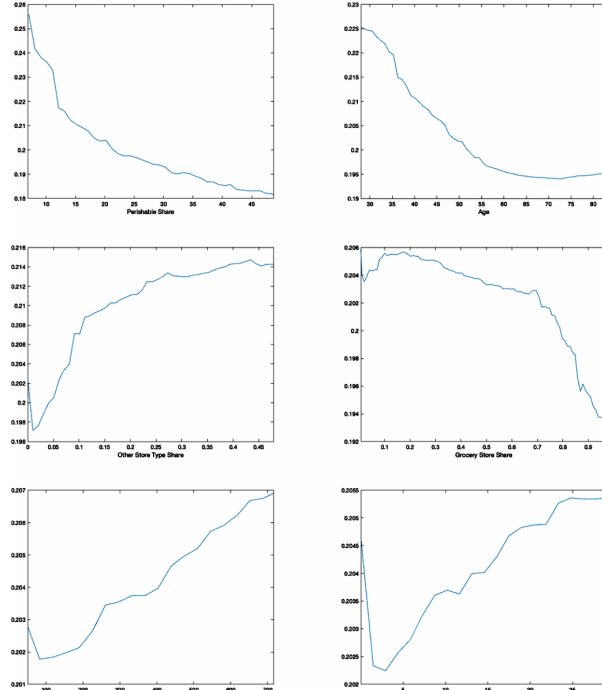


Figure 6: partial dependence profiles

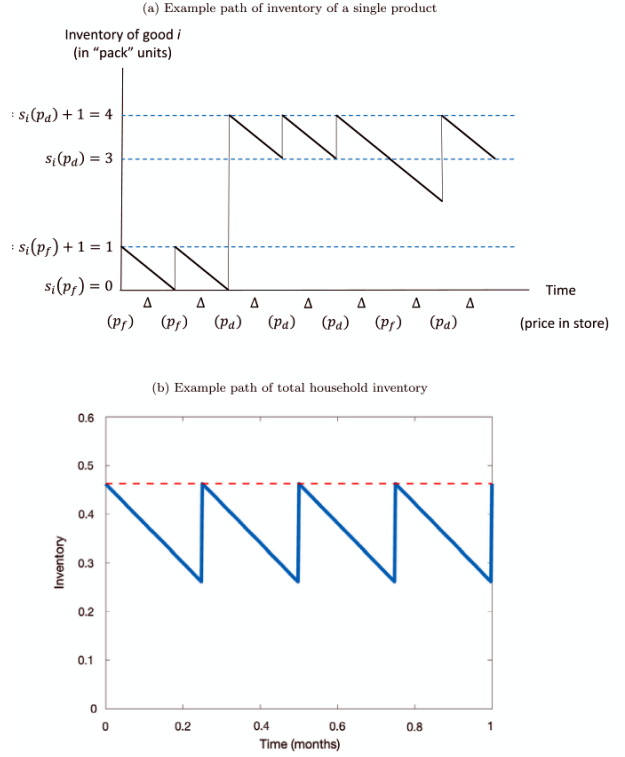


Figure 7: example inventory paths

Read:

- Figure 6 shows partial dependence profiles for the six most important predictors.
- Inventory ratios decline with perishability and age patterns are non-linear.
- Store type and house prices have important effects on predicted inventory ratios.
- Figure 7 illustrates the model's inventory path for individual goods and aggregate household inventory.
- The household stockpiles after low-price events and then gradually draws inventory down between trips.

Economic message: The empirical heterogeneity motivates the model: inventory is shaped by perishability, purchase opportunities, and working-capital constraints.

5.5 Table 3 + Figure 8: Calibration

Table 3
Calibration of model parameters.

Name		Perishability Group (i)				Source/target
		1	2	3	4	
Depreciation rate	δ_i	4.29	1.76	0.34	0.00	See Appendix C
Shelf life (in months)	t_i	Δ^*	0.39	2.06	27.92	
Consumption flow (in %)	C_i	12.47	11.52	12.90	63.1	Match NCP expenditure shares.
Deal probability	s_i	0.21	0.30	0.29	0.27	Match NCP price moments.
Full price	$p_{f,i}$	1.07	1.07	1.08	1.08	Match NCP price moments and $E[p_i] = 1$.
Deal price	$p_{d,i}$	0.74	0.83	0.80	0.78	
Regression coefficients	$\hat{\alpha}_{ij}$	0.77	0.82	0.84	0.81	Estimation of (26) by WLS.
	$\hat{\beta}_{ij}$	3.00	0.94	4.41	0.77	
Bulk discount function	α_i	0.78	0.81	0.82	0.77	α_i, β_i match relation between NCP pack size and unit price.
	β_i	3.05	0.92	4.30	0.73	
	$\bar{\beta}_i$	2.63	1.56	3.18	1.16	
Trip fixed cost	k	0.0139				Match NCP trip interval.

Notes: This table shows the model parameters by perishability group. Group 1 contains the most perishable products and Group 4 contains the least perishable. The calibration approach is described in Section 4.4. In the model, one unit of a product from group i stored for a period of t months since purchase provides consumption of $e^{-\delta_i t}$ units if $t < t_i$ and 0 units if $t \geq t_i$. C_i is the % of total Nielsen/Q spending accounted for by each group. s_i is the probability of a sale. $p_{f,i}$ is the price in the event of a sale, and $p_{d,i}$ is the full price. $\hat{\alpha}_{ij}$ and $\hat{\beta}_{ij}$ are weighted least squares coefficients of regression equation (26), estimated separately for each group. In the model, we normalize the price of the standard pack size to one, and the price of other pack sizes reflect percentage deviations from the standard pack size. We therefore set $\alpha_i = \frac{\beta_i}{1 + \beta_i \bar{\beta}_i}$ and $\beta_i = \frac{\bar{\beta}_i}{1 + \beta_i \bar{\beta}_i}$. $\bar{\beta}_i$ is chosen to maximize the within- R^2 of (26). Note, because the implied shelf life for group 1, t_1 , is somewhat lower than the data trip interval (which we calibrate k to match), we set $t_1 = \Delta^*$, the household's optimal trip interval. This effectively means that products from group 1 cannot be stockpiled.

Table 3: calibration of model parameters

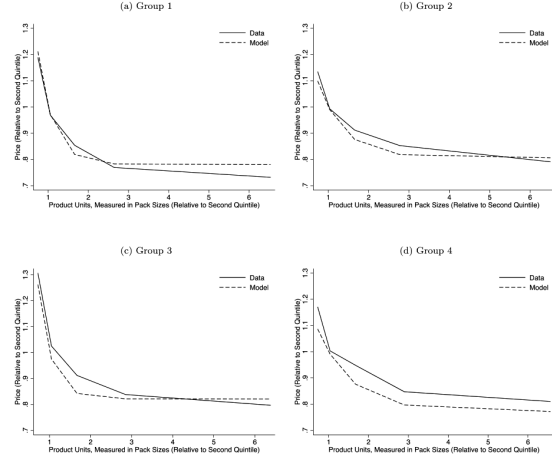


Figure 8: bulk calibration

Read:

- Table 3 calibrates four perishability groups, from very perishable to very durable goods.
- The least perishable group accounts for the largest expenditure share.
- Deal probabilities, deal prices, full prices, depreciation rates, shelf lives, bulk parameters, and trip fixed costs are matched to scanner data moments.
- Figure 8 shows that the model reproduces the relationship between package size and unit price within product groups.

Economic message: The model is not estimated from returns. It is calibrated using observable shopping, price, bulk, and durability moments, then used to infer financial returns.

5.6 Table 4 + Figure 9: Financial returns to working capital

Table 4
Financial returns to household inventory investment.

Work. Cap. (I , % cons.)	Inventory (% spend.)	Cash (% I)	% Savings	Interval Δ^* (months)	$r'_{L,t}$ (trips)	Return (%)	
			Deal	Bulk		Marg.	Avg.
2.5	1.8	35.1	6.2	12.8	0.23	2	215.2
5.0	4.7	19.0	9.9	15.7	0.26	5	56.6
7.5	7.4	13.6	10.8	17.3	0.28	7	28.6
10.0	10.7	9.4	11.8	16.1	0.27	10	13.7
12.5	13.6	7.8	11.9	16.7	0.28	12	6.3
15.0	16.5	6.6	12.0	17.1	0.28	14	3.0
17.5	19.4	5.7	12.1	17.2	0.28	16	1.6
20.0	22.1	5.0	12.1	17.2	0.28	18	0.8
22.5	24.9	4.4	12.1	17.2	0.28	20	0.5
25.0	27.6	4.0	12.2	17.2	0.28	22	0.2
27.5	30.3	3.6	12.2	17.2	0.28	24	0.1
30.0	34.3	3.3	12.2	17.2	0.28	27	0.1
32.5	37.2	3.1	12.2	17.2	0.28	29	0.0
35.0	39.9	2.9	12.2	17.2	0.28	31	0.0

Notes: This table is constructed by solving the model for different values of the working capital constraint I , increasing it by 2.5% of (exogenous) annual consumption in each row. The working capital ratio in column 1 is available working capital I , expressed as a percentage of annual consumption. Column 2 shows the value of inventory immediately prior to a trip as a percentage of total annual spending. Column 3 shows the annual average value of cash set aside to pay for the next trip as a share of annual total working capital. The remaining share of working capital is invested in inventory. Columns 4 and 5 show in-store savings achieved in % of base spending, which is annual spending assuming no stockpiling ('untargeted' or 'inattentive' shopping, $s_{ij} = 0$) and the trip interval is the interval associated with purchasing the standard pack size of each product. Deal savings are in-store savings due to buying an item on sale. Bulk savings are in-store savings due to buying a larger pack size. Column 6 shows the length of time of the optimal interval Δ^* between trips, measured in months. $r'_{L,t}$ in column 7 is the optimal deal-shopping strategy for goods with a shelf life of at least six months (group 4), expressed in the number of trips the household is willing to purchase the product in advance of consuming it when the product is on sale. The financial returns in columns 8 and 9 incorporate not only in-store savings but also depreciation and trip fixed costs. The average return is computed relative to a working capital benchmark of 2.5% of annual consumption.

Table 4: financial returns to household inventory investment

Read:

- Table 4 shows very high marginal returns at low levels of working capital.
- Marginal returns decline quickly as working capital rises and additional inventory opportunities are exhausted.
- At empirically relevant inventory levels, average net returns remain economically large.
- Figure 9 shows that adding household working capital dramatically raises average portfolio returns for low-income and low-asset households.
- In the bottom asset quintile, including working capital raises the average return from near zero to about 45.5%.

Economic message: Inventory management can dominate standard financial returns for households with little wealth. This helps explain why scarce liquidity may be held in goods rather than financial assets.

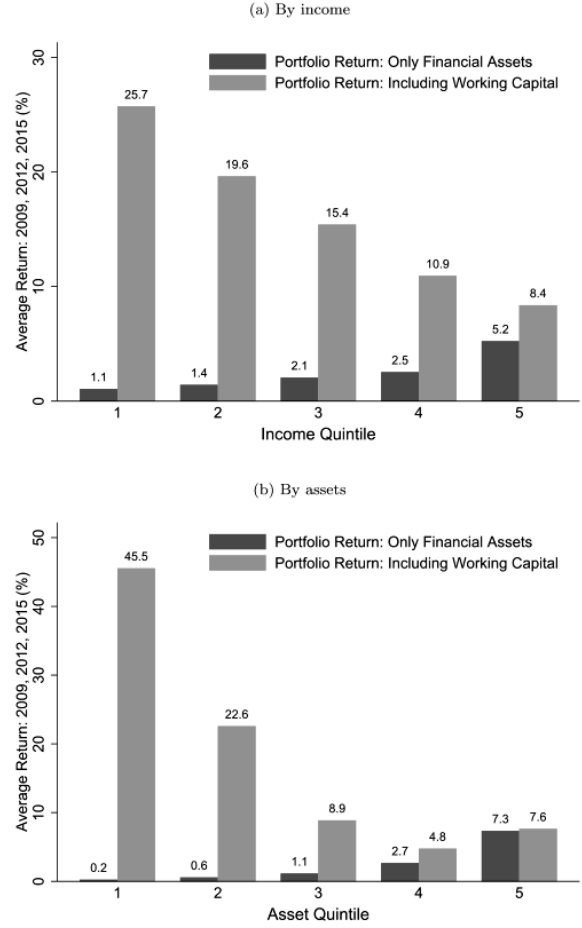
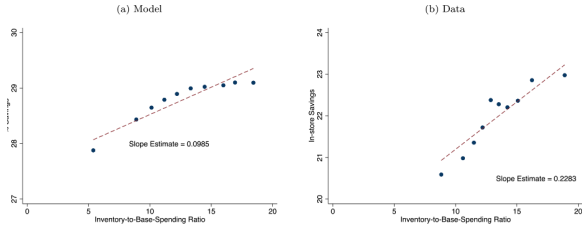


Fig. 9. Implied effect of working capital on average returns.

Figure 9: portfolio returns including working capital

5.7 Figure 10 + Figure 11: Gross returns and mover evidence



Gross return: in-store savings and inventory.
construct Panel (a) we vary working capital (I) and compute the in-store savings and inventory as a share of spending. We plot the relationship between in-store savings and inventory on the number of trips for comparability with the data. Panel (b) uses the NCP over 2013 and 2014 to illustrate the relationship between in-store savings and inventory.

Figure 10: in-store savings and inventory

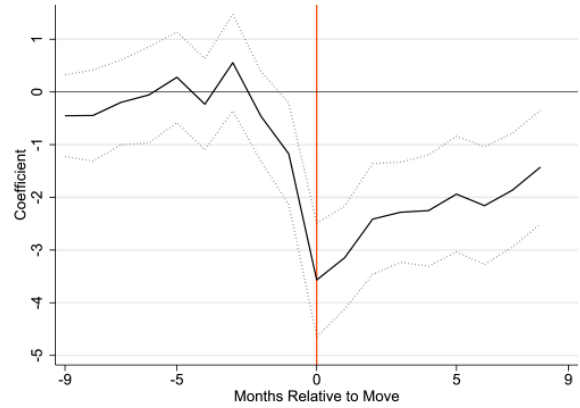


Fig. 11. Deal transactions around move dates.

Notes: This figure shows the percentage point change in the share of purchases associated with a coupon or self-reported deal. For households who move to a new location, the move date is the date of the move.

Figure 11: deal transactions around move dates

Read:

- Figure 10 shows a positive relationship between inventory and in-store savings in both the model and the data.
- The data slope is larger than the model slope, suggesting that observed inventory variation is associated with substantial gross savings.
- Figure 11 shows that deal transactions fall around moves.
- The estimated drop in deal transactions is about 3.6 percentage points in the move month.

Economic message: Households with more inventory obtain more in-store savings, and movers reduce deal-based stockpiling when carrying inventory becomes costly.

5.8 Figure 12 + Figure 13: Deal savings and bulk savings

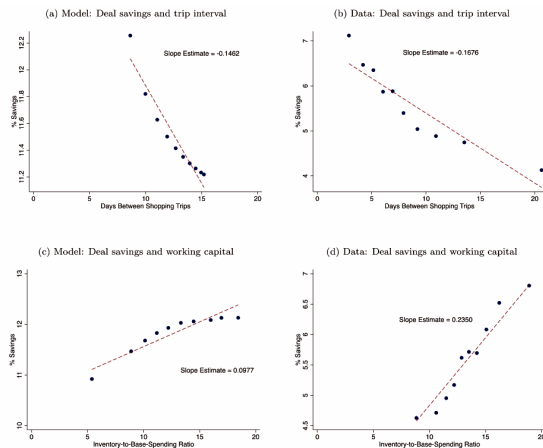


Figure 12: deal savings, trip interval, and working capital

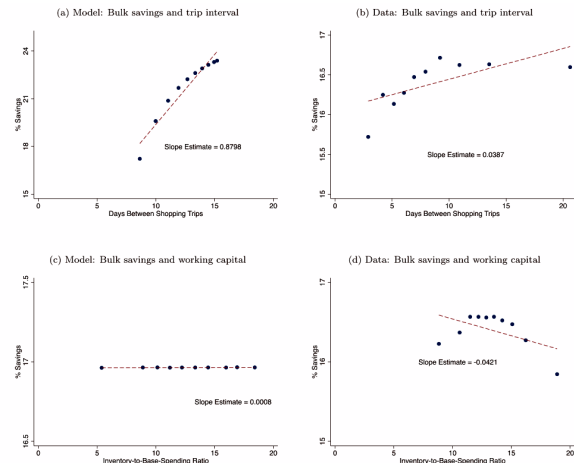


Figure 13: bulk savings, trip interval, and working capital

Read:

- Figure 12 shows that deal savings decline with the trip interval and rise with working capital.

- The signs match the model: frequent shoppers can exploit deals, while more working capital supports stockpiling.
- Figure 13 shows that bulk savings rise with trip interval in the model and only weakly in the data.
- Bulk savings are almost flat in working capital in the model and slightly negative in the data.

Economic message: Deal savings are closely tied to household working capital. Bulk savings are more closely tied to trip frequency and fixed shopping costs.

6 Short summary

- The paper measures household consumer-goods inventory using NielsenIQ scanner data.
- The average household holds inventory worth hundreds of dollars and about two to three months of scanner-measured spending.
- Inventory is especially important for low-income and low-asset households.
- Shelf life and moving behavior validate the inventory measure.
- The model treats inventory as working capital used to lower the cost of a fixed consumption stream.
- Households earn returns by buying on deals and by using larger package sizes with lower unit prices.
- Working capital is scarce, so marginal returns are high at low levels of inventory and decline as the household accumulates more stock.
- The model-implied returns are high enough to matter for portfolio choice.
- Including household inventory raises average portfolio returns sharply for low-income and low-wealth households.
- Reduced-form evidence on in-store savings, movers, deal transactions, and bulk savings supports the model mechanism.
- The paper suggests that household inventory management may help explain why some households hold little financial wealth or avoid risky financial assets.

7 Conclusion

7.1 Core conclusion

Baker, Johnson, and Kueng show that household inventory management can generate substantial financial returns. Consumer goods inventories are not merely unused groceries or household clutter; they are a form of working capital that households use to save money on recurring consumption.

7.2 Economic intuition

The intuition is straightforward. A household with extra cash can buy goods when they are on sale or buy larger packages with lower unit prices. The benefit is a lower cost of supplying the same consumption flow. The cost is that the household must tie up cash in illiquid inventory and bear depreciation, storage, and trip-related costs. For low-wealth households, the first dollars devoted to working capital can earn very high returns.

7.3 Takeaway

The paper expands household finance by treating consumer goods inventory as a portfolio asset. It shows that nonfinancial working capital can offer high, declining, and illiquid returns. This helps explain why households

with limited liquidity may rationally prioritize inventory management over participation in financial markets.